

Dureksha Arangala

Undergraduate in Data Science

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PROFESSIONAL SUMMARY

Final-year BSc (Hons) Data Science undergraduate at SLTC with a strong passion for machine learning, artificial intelligence, statistical analysis, and data analytics. Skilled in extracting insights from complex datasets, developing predictive models, and communicating findings through data visualization. Eager to contribute analytical thinking, problem-solving skills, and data-driven solutions that support informed business decisions and organizational growth.

TECHNICAL SKILLS

Programming: Python, SQL

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Microsoft Excel

ML & DL: Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, NLTK, spaCy, Transformer Models

Databases: PostgreSQL, MySQL, MongoDB

Big Data & Data Engineering: PySpark, Apache Kafka, ETL Pipelines, Data Warehousing

MLOps & Development Tools: DagsHub, GitHub, Docker, DVC, Apache Airflow, Jupyter Notebook, Visual Studio Code, Google Colab

PROJECTS

- **Liya – Postpartum Depression Prediction & Recovery App (FYP Ongoing)**
Developing an end-to-end AI-powered mobile health application for early postpartum depression prediction using Machine Learning, Explainable AI (SHAP), and Conversational AI. Integrating predictive analytics, personalized recovery recommendations, and continuous mental health monitoring through a Flutter-based mobile application.
Tools: Python, Scikit-learn, TensorFlow, SHAP, Flutter, Supabase, PostgreSQL, Google Dialogflow CX
- **Telco Customer Churn Prediction – End-to-End ML & MLOps System**
Built an end-to-end MLOps pipeline to predict telecom customer churn, automating data processing, model training, experiment tracking, and deployment. Implemented version-controlled ML workflows with automated pipelines and deployed a real-time churn prediction API using Docker.
Tools: Python, Scikit-learn, Pandas, NumPy, DVC, DAGsHub, MLflow, Apache Airflow, FastAPI, Docker, DagsHub
- **E-Commerce Big Data Analytics Pipeline**
Developed an end-to-end big data analytics pipeline to process, transform, and analyze large-scale e-commerce datasets. Implemented data ingestion, ETL processing, and analytics workflows using distributed data processing technologies.
Tools: Python, Pandas, NumPy, PySpark, MongoDB Atlas, Google Colab, Jupyter Notebook, GitHub
- **End-to-End NLP Hate Speech Classification**
Developed an NLP model for automated hate speech detection and deployment.
Tools: Python, Pandas, NumPy, NLTK, Regex, TensorFlow, Keras, Scikit-learn, FastAPI, Uvicorn, Jinja2, GitHub, Jupyter Notebook, Docker.
- **Customer Demand Forecasting System**
Built a machine learning forecasting model for demand prediction and inventory planning.

Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Time Series Analysis, Feature Engineering, Machine Learning Models, Jupyter Notebook, VScode, GitHub.

EDUCATION

Sri Lanka Technology Campus

B.Sc (Hons) Data Science
Expected Graduation: 2027
Current GPA: 3.73

GCE Advanced Level (A/L) | 2021

WP/Jaya/P/Dharmapala Vidyalaya
Biological Science Stream (Biology, Chemistry, Physics)

CERTIFICATION & ACHIEVEMENTS

- Data Science 101 – IBM SkillsBuild
- InspiHER{Tech} V4.0 – HackNova Competition | First Runner-Up SLTC
- IEEE Xtreme Programming Competition 19.0 – Participation Certificate
- BioFusion Hackathon – Participation Certificate | IEEE Engineering in Medicine and Biology Society (EMBS)
- Emerging Scholars' Research Programme – Certificate Phase Completion | Faculty of Postgraduate Studies & Research, SLTC

LEADERSHIP & VOLUNTEER EXPERIENCE

- **Advisory Panel(Secretary)- IEEE Computer Society Student Branch Chapter of SLTC | 2025-2026**
Managed chapter documentation, coordinated executive activities, and supported GitGenius'26, CodeMania v6.0, Lumora v5.0, and From Awareness to Action.
- **Editorial Deputy Head (Subcommittee) - IEEE CS SBC of SLTC | 2024 – 2025**
Led editorial activities by reviewing and coordinating promotional content for chapter events.
- **Deputy Head – Logistics Team | ITIA Career v1.0**
Coordinated logistics and operational activities to ensure successful event execution.
- **Secretary Team Member | MASTER DESIGNER v3.0 | Media Unit, SLTC**
Supported event documentation, coordination, and administrative operations.
- **Content & Caption Team Head | Sipsari Yathra | LEO Club, SLTC**
Led content creation and social media communications to promote the project.

SOFT SKILLS

- **Problem Solving** - Developing effective solutions to complex technical and real-world challenges.
- **Analytical Thinking** - Analyzing information, identifying patterns, and making data-driven decisions.
- **Teamwork & Leadership** - Collaborating with teams, coordinating activities, and supporting project success.
- **Project Management** - Planning, organizing, and managing tasks to deliver projects within deadlines.
- **Adaptability & Continuous Learning** - Quickly learning new technologies and continuously improving technical skills.

REFERENCES (AVAILABLE UPON REQUEST)

Dr. Chameera De Silva

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